

SMARTBRIDGE — Solution Requirements

Functional & Non-Functional Requirements for MedSafe AI

Date:	August 12, 2026	Team ID:	MEDSAFE_AI_01
Project Name:	MedSafe AI — Healthcare Safety & Preventive Awareness Platform	Max Marks:	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Local session-based access without requiring external public cloud login to protect health privacy.	Medium
2	Authorization levels	Standard Patient/Caregiver view for uploads/queries; Admin view for updating drug interaction JSON rules.	Low
3	External interfaces	Connects to local Ollama (LLaMA 3 instance) via REST API and PyTesseract OCR engine locally.	High
4	Transactions processing	Processes image-to-text OCR parsing, fuzzy string matching, and symptom triage score calculation.	High
5	Reporting	Generates printable/downloadable Patient Safety Reports detailing dosage, interaction risk, and triage level.	High
6	Business rules / Logic	If red-flag keywords (e.g. chest pain) are detected, override AI advice and display immediate ER Alert banner.	High
7	Compliance & Privacy	Ensures zero medical data leaves local client machine (HIPAA/GDPR compliant local execution).	High
8	Typo Resilience	Fuzzy string matching resilience (RapidFuzz threshold >80%) for drug names entered by users.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast deterministic drug lookup and responsive OCR extraction.	Interaction check < 1 sec; OCR extraction < 4 sec.
2	Scalability	Support local multi-threading for batch prescription processing.	Handles up to 10 page documents concurrently.
3	Security & Data Privacy	Complete air-gapped local execution without transmitting PHI to third-party APIs.	100% local processing; 0 external data calls.
4	Reliability & Availability	Fallback to local rule database if LLaMA 3 model instance fails or times out.	99.9% system availability locally.
5	Usability & Accessibility	High-contrast, large-font web interface suitable for senior patients and caregivers.	WCAG 2.1 AA compliant UI layout.
6	Portability	Deployable as a standalone cross-platform Docker container or Streamlit app.	Runs on Windows, macOS, and Linux.